Connectivity-defined coordinates and dynamical motifs in multitask recurrent networks: methods and results

Working report

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Abstract

This report documents the methods and numerical results of an analysis of trained multitask recurrent neural networks (RNNs). We identify approximate equitable partitions of the recurrent weight matrix, quantify how far the resulting group-averaged ("quotient") coordinates capture the recurrent dynamics, locate the networks' fixed-point motifs across task epochs, and relate the motif directions to the connectivity coordinates. The principal model set is ten 256-unit networks (three released, seven trained for this study); other network sizes appear only through their released seeds. Results are reported across a sweep of the group number k rather than at a single operating scale. Every comparison is reported against its size-matched null and, where computed, against an activity-defined partition. Positive, negative, and null outcomes are reported with equal prominence. The unit of replication is the trained network (seed).

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1 Model set and training

1.1 Architecture and task set

We analysed leaky discrete-time RNNs with softplus activation $\sigma(x) = \log(1 + e^x)$ and diagonal recurrent initialisation, matching the primary model class of Driscoll, Shenoy & Sussillo (2024). The

recurrent state $h(t) \in \mathbb{R}^N$ evolves as

$$h(t{+}1) = (1-\alpha)\,h(t) + \alpha\,\sigma(W_{\text{rec}}h(t) + W_{\text{in}}u(t) + b_{\text{rec}})\,, \qquad z(t) = W_{\text{out}}h(t) + b_{\text{out}},$$

with leak $\alpha = 0.2$ (recovered exactly from stored Jacobians; §8). Inputs $u(t)$ comprise one fixation channel, four stimulus channels (two circular modalities as sine/cosine pairs), and 15 one-hot rule channels (input dimension 20); outputs comprise one fixation and two response-direction channels. The task set is the 15 cognitive tasks of the multitask paradigm. All hidden-activity analyses used noiseless test trials.

1.2 Networks analysed

The principal analyses use ten independently trained 256-unit networks (seeds 0–9): seeds 0–2 are the networks released by Driscoll et al.; seeds 3–9 were trained for this study on the corrected objective (below) and have L1 activity/weight coefficients 0, L2 coefficients 10^{-6}, and 212–220 active units (task-epoch activity variance $> 10^{-3}$). Mean accuracy across tasks is $\approx$ 0.97–0.99 (released) and $\approx$ 0.97 (retrained; per-task range: non-decision tasks 0.998–1.000, decision-making tasks 0.94–0.96). Networks at 128, 512 and 1024 units are represented only by their released seeds 0–2 and are used solely for size comparisons. An earlier batch of retrained seeds was produced under an unintended regularisation coefficient and has been superseded; that batch and its outputs are documented in the companion report `legacy_uncorrected_seeds_report`.

2 Structural partition identification

Analyses were restricted to active units. The recurrent block was split into $W^+ = \max(W, 0)$, $W^- = \max(-W, 0)$ with $W = W_{\text{rec}}^T$ (each unit characterised by the signed pattern of recurrent inputs it receives). A partition into k groups is an indicator matrix $H \in \{0,1\}^{n\times k}$; equitability is scored by

$$\Gamma_{\text{signed}}(H) = \|W^+H - HD^{-1}H^TW^+H\|_F + \|W^-H - HD^{-1}H^TW^-H\|_F, \qquad D = H^TH,$$

minimised by an approximate signed Weisfeiler–Leman refinement (k-means reclustering of the concatenated profile $[W^+H \,|\, W^-H]$; 20 iterations, 20 initialisations). The full k-sweep (k = 3–60) was cached, together with an independently computed activity partition (Ward clustering of normalised task-epoch variance profiles) at each k.

Reporting scale. Results below are reported across a sweep $k \in \{8, 12, 16, 20, 25, 30, 40, 50\}$. A single "operating" scale of 20% of the active-unit count (giving k = 33–51 for the 256-unit networks) is used only where one scale is required; it is not the basis of any conclusion, and the k-sweep is the primary presentation.

Group-size distribution and singletons. The refinement does not constrain group sizes; groups are unequal and some are singletons. Median over the ten 256-unit networks of the connectivity partition (of $\approx$ 210 active units): at $k = 8$, group sizes 1/26/63 (min/median/max) with 1 singleton; at $k = 20$, 1/9/26 with 2 singletons; at $k = 30$, 1/6/25 with 5 singletons; at $k = 50$, 1/3/18 with 14 singletons. Singleton groups are retained (they carry a one-unit row of the quotient); no size floor is imposed. Because every comparison partition (random, activity, spectral, modularity) is matched to the same group-size multiset (§20), the effect of the size distribution is controlled rather than assumed away.

3 Lumpability of the linear recurrent map (Result 1)

Let $P = H(H^T H)^{-1} H^T$. For each leading eigenvector v of a matrix M we measured the leakage $\|v - Pv\|/\|v\|$, step $R^2 = 1 - \|v - Pv\|^2/\|v - \bar{v}\|^2$, and the subspace invariance $\|(I-P)MP\|/\|MP\|$, each standardised against size-matched random partitions (200 permutations) and reported as a signed *score* (positive-for-more-structure). $M = W_{\text{rec}}^T$ and the raw state-update matrix were both evaluated.

3.1 Results (k-sweep, ten 256-unit networks)

k	leak score (W^T)	#>0	leak score (W)	invariance score	#>0 (inv)
8	+27.8	10/10	+21.9	+1.1	7/10
12	+29.8	10/10	+22.6	+2.8	7/10
16	+28.4	10/10	+25.3	+5.1	8/10
20	+27.8	10/10	+21.2	+7.0	9/10
25	+22.3	10/10	+20.0	+9.4	9/10
30	+21.2	10/10	+18.1	+11.7	10/10
40	+20.3	10/10	+14.7	+16.8	10/10
50	+19.4	10/10	+12.8	+17.7	10/10

The eigenvector-leakage score was positive in 10/10 networks at every k; the subspace-invariance score rose with k and was positive in 7–10/10 (weaker at small k). For the canonical network (seed 1, $k = 51$) a leading eigenvector had step $R^2 = 0.89$ on the recovered groups versus $R^2 = 0.20$ under a size-matched random regrouping. For the released seeds 0–2 at other sizes (operating scale), the leakage score was +12.9 (128), +22.6 (512), +26.8 (1024).

3.2 Degree/training surrogate control

The identical pipeline (WL optimisation at the operating k, then equitability z and step R^2) was applied to three structure-destroying surrogates: (i) a row shuffle preserving each unit's incoming-weight strength; (ii) a full shuffle; (iii) an i.i.d. Gaussian matched in Frobenius norm. For the ten corrected 256-unit networks:

matrix	equitability score	step R^2
trained	+7.35	0.784
row shuffle	+12.26	0.693
full shuffle	+3.14	0.415
matched Gaussian	+2.12	0.275

The trained matrix exceeded the row shuffle on step R^2 in 10/10 networks and on equitability score in 0/10 (the row shuffle is more equitable than the trained matrix in every network). Equitability magnitude is thus not higher for the trained matrix than for a strength-preserving shuffle; step R^2 is.

The reason the row shuffle scores *higher* on equitability is a property of the null, not of trained structure. The equitability score is $z = (\overline{\Gamma_{\text{null}}} - \Gamma_{\text{obs}})/\text{sd}(\Gamma_{\text{null}})$, where Γ_{null} is recomputed from each surrogate's own size-matched random partitions on that surrogate's own matrix (each surrogate is scored against its own null, verified in code). On a representative network (seed 0, $k = 20$): the trained matrix gave $\Gamma_{\text{obs}} = 157.5$ against its own null mean 151.8 (sd 2.5), i.e. $z = -2.25$ — the WL partition of the trained matrix is *not* more equitable than a random same-size partition of it. The row-shuffled matrix gave $\Gamma_{\text{obs}} = 128.0$ against its own null mean 161.9 (sd 1.7), i.e.

$z = +19.5$. Within-row shuffling makes rows exchangeable and separable essentially by incoming strength (degree); WL can then group rows to drive Γ far below a degree-blind random null, whereas the trained matrix's structure is not of that degree-separable kind. The equitability z thus rewards degree separability, which the shuffle amplifies; this is why the load-bearing structural claim is lumpability (training-made, trained > shuffle 10/10; §3) rather than equitability magnitude.

4 Partition non-identifiability and subspace stability

Re-running the refinement from 20 initialisations (256-unit networks): the cost Γ_{signed} was reproducible (coefficient of variation $\approx 3\%$) while the specific assignments were not (mean pairwise adjusted Rand index ≈ 0.40). The group-constant subspaces of restart pairs had mean squared principal-angle cosine ≈ 0.63 versus ≈ 0.24 for size-matched random partitions (mean principal angle $\approx 34°$). Across-restart coefficients of variation were $\approx 2.5\%$ (step-R^2) and $\approx 10\%$ (leakage). Single-partition analyses used a co-association consensus.

5 Comparison to spectral and modularity partitions

A control for whether the backbone results require the signed-WL partition specifically, or would follow from any sensible graph partition at matched k. At each k we built, on the same active recurrent block, a normalised spectral-clustering partition (affinity $|W| + |W|^T$) and a greedy modularity partition (same weighted graph, forced to k communities), and evaluated the three label-vector metrics behind Results 1/3/4: eigenvector leakage (Result 1, lower is more lumpable), median motif-direction containment (Result 3, higher), and top-3 quotient-mode energy (Result 4, higher). Medians over the ten 256-unit networks; signed-rank compares WL to each alternative.

metric	k	WL	spectral	modularity	random	WL vs spec / mod (p)
leakage ↓	12	0.656	0.806	0.856	0.868	0.002 / 0.002
	20	0.644	0.795	0.855	0.863	0.002 / 0.002
	30	0.635	0.751	0.842	0.830	0.002 / 0.002
containment ↑	12	0.470	0.222	0.124	0.222	0.002 / 0.002
	20	0.598	0.285	0.166	0.294	0.002 / 0.002
	30	0.677	0.367	0.245	0.365	0.002 / 0.002
top-3 ↑	12	0.802	0.511	0.623	0.547	0.002 / 0.002
	20	0.641	0.360	0.503	0.426	0.002 / 0.004
	30	0.560	0.318	0.448	0.348	0.002 / 0.002

The signed-WL partition exceeded both spectral and modularity partitions on all three metrics at every k (10/10 networks, $p \leq 0.004$). The spectral and modularity partitions were at or below the size-matched random floor on containment (modularity below random, since dense unsigned communities are not aligned with the motif subspace). The lumpable/containing/concentrating coordinates therefore come specifically from the signed-equitability partition, not from graph community structure in general. (The leakage advantage is partly definitional — WL minimises an equitability cost tied to eigenvector leakage — but containment and top-3 energy are motif quantities that WL does not optimise, and it wins there too.)

6 Fixed-point motifs

For each task epoch, holding the epoch input constant, fixed and slow points of the autonomous dynamics were located by gradient descent on the velocity $q(h) = \frac{1}{2}\|-h + \sigma(W_{\text{rec}}h + W_{\text{in}}u + b)\|^2$

(thresholds $q < 10^{-8}$ for isolated fixed points, $q < 10^{-6}$ for the slow cloud). Analysed epochs were fixation, stimulus (`stim1/2`) and delay (`delay1/2`); the pure response epoch (`go1`) was analysed separately (§11). Each epoch was classified from local Jacobian spectra as a point attractor (all $|\lambda| < 1$), a saddle (≥ 1 mode $|\lambda| > 1$), or a line/ring attractor (marginal mode, locally one-dimensional cloud); a rotational motif would show a Jacobian eigenvalue with imaginary part > 0.05.

6.1 Taxonomy (corrected models, all ten seeds)

Across 490 classified task epochs the inventory was 263 line attractors, 101 point attractors, 76 saddles, 50 mixed, and 0 rotations; all fixed-point Jacobian spectra were near-real. Epoch-type gate counts (matched / total): pooled over seeds 0–2, memory delays $\rightarrow$ line attractor 40/48, decision stimulus $\rightarrow \geq 1$ saddle 30/42, sensory stimulus $\rightarrow$ point attractor 9/12; pooled over all ten seeds, 130/160, 109/140, and 13/40. Each motif was summarised by a unit direction p: the marginal (memory) mode of a line attractor or the leading unstable (decision) axis of a saddle.

7 Motif directions in the connectivity subspace (Result 3, containment)

We measured the fraction of each motif direction inside the group-constant subspace, $\|Pp\|/\|p\|$, and at matched k the excess containment $\Delta = f_{\text{obs}} - \overline{f_{\text{rand}}}$ for the connectivity and activity partitions (each minus its own size baseline). Across the ten 256-unit networks:

k	Δ_{role}	Δ_{activity}	z (role−act)	role > act
8	0.121	0.016	+4.01	10/10
12	0.183	0.010	+4.97	10/10
16	0.239	0.014	+6.35	10/10
20	0.280	0.012	+7.31	10/10
25	0.295	0.021	+6.24	10/10
30	0.282	0.020	+6.04	10/10
40	0.268	0.018	+6.03	10/10
50	0.270	0.037	+5.13	10/10

The connectivity excess containment exceeded the activity partition's in 10/10 networks at every k; the activity excess was near zero throughout.

7.1 W_{rec}-eigenbasis null (strong null; negative for containment)

The random and activity partitions are weak nulls. A stronger, connectivity-derived null is the eigenbasis of W_{rec} itself: the span of the top-k eigenvectors of W_{rec} (by $|\lambda|$, real parts, orthonormalised) matched in dimension to the k-group constant subspace. We compared the raw containment $\|Pp\|/\|p\|$ of the motif directions onto each subspace (median over the ten networks; two-sided signed-rank):

k	containment (conn)	containment (W_{rec}-eig)	conn > eig (motif frac.)	Wilcoxon p
8	0.320	0.678	0.00	0.002
12	0.470	0.706	0.00	0.002
16	0.540	0.716	0.02	0.002
20	0.598	0.729	0.03	0.002
25	0.652	0.743	0.04	0.004
30	0.677	0.757	0.08	0.002
40	0.716	0.776	0.19	0.014
50	0.769	0.797	0.23	0.160

The top-k W_{rec} eigenbasis contained the motif directions more than the connectivity group-constant subspace at every k (signed-rank in the eigenbasis's favour up to $k = 40$; per-motif the connectivity subspace won in 0–23% of motifs). The connectivity partition is therefore *not* a privileged k-dimensional subspace for the motif directions relative to the leading eigenvectors of the same matrix; its containment advantage in §7 is specifically over the activity partition and size-matched random partitions, not over the W_{rec} spectrum. This is expected — the leading eigenvectors are by construction the best k-dimensional linear fit to the slow directions the motifs align with — but it bounds the containment claim and is stated because it is the first null a reader is likely to name.

A second natural k-dimensional subspace, defined by the task inputs rather than by recurrence, is the span of the top-k right-singular directions of W_{in} over the active units. It too contained the motif directions more than the connectivity groups at every k (median containment 0.75 vs. 0.60 at $k = 20$; connectivity above the input span in 1–13% of motifs; signed-rank $p = 0.002$ at all k). The connectivity partition's containment advantage is therefore specifically over the *activity* and size-matched *random* partitions (§7); it is not an advantage over either the leading recurrent eigenvectors or the input-weight span.

8 Quotient reduced model (Result 2)

At a fixed point $J = (1 - \alpha)I + \alpha\,\mathrm{diag}(g)\,W_{\text{rec}}$ with $g = \sigma'(x^*)$; $\alpha = 0.200$ and the per-unit gains g are recovered exactly ($\sim 10^{-8}$) from a stored Jacobian and W_{rec}, with no fitting. We formed $Q_{\text{struct}} = D^{-1}H^T W_{\text{rec}} H$, $Q_{\text{pred}} = (1 - \alpha)I + \alpha\,\mathrm{diag}(\bar{g})\,Q_{\text{struct}}$, and $Q_{\text{task}} = D^{-1}H^T \bar{J} H$. The top-$m$ eigenvectors of Q, lifted to units and orthonormalised, form B; for a motif direction v, $A_{\text{raw}} = \|B^T v\|/\|v\|$ and $A_{\text{cond}} = \|B^T v\|/\|P_H v\|$. Nulls are size-matched random partitions with their own recomputed quotient, plus the activity partition at matched k.

8.1 Prediction across the k-sweep (Q_{pred}, $r = 3$, ten networks)

k	A_{cond} (conn)	random	conn > rand	z
8	0.932	0.736	10/10	+2.62
12	0.890	0.660	10/10	+3.56
16	0.874	0.625	10/10	+3.88
20	0.816	0.612	10/10	+3.45
25	0.809	0.598	10/10	+3.32
30	0.797	0.595	10/10	+3.42
40	0.783	0.594	10/10	+3.37
50	0.787	0.611	10/10	+3.07

Connectivity exceeded its random-quotient null in 10/10 networks at every k. Eigenvalue agreement at $k = 20$: $|\lambda_Q - \lambda_J| = 0.046$ (connectivity) vs. 0.191 (random) and 0.141 (activity).

8.2 Compression (A_{raw}, $k = 20$, $r = 3$, median over ten networks)

Quotient Q_{pred} ($k^2 + k = 420$ free parameters) $= 0.479$; full mean Jacobian ($\approx 4 \times 10^4$ parameters) $= 0.801$; full W_{rec} top-3 eigenbasis $= 0.466$; W_{rec} SVD basis $= 0.113$; random directions in the group subspace $= 0.205$; containment ceiling $\|P_H p\| = 0.578$.

8.3 Gain ablations (A_{cond}, $k = 20$, $r = 3$, median over ten networks)

predictor	A_{cond}
Q_{pred} (group-mean gains)	0.816
scalar gain (one global)	0.749
permuted group gains	0.680
no gains (Q_{struct})	0.670
cross-condition gains	0.775
gains only (no quotient)	0.460
size-matched random (null)	0.621

The gains-only predictor falls below the random-partition null; the no-gain quotient exceeds it by 0.049.

8.4 Eigenvalue-matched non-motif control (negative)

Q_{pred}'s prediction of the motif direction v was compared with its prediction of a non-motif eigenvector q of the *same* mean Jacobian, chosen with $|\lambda_q|$ nearest $|\lambda_v|$ and alignment to $v < 0.30$; partition, gains, r and conditioning identical. Across the ten networks the fraction of conditions in which the motif direction was predicted better than the eigenvalue-matched non-motif direction was: $k = 8$, 54%; $k = 12$, 52%; $k = 16$, 50%; $k = 20$, 46%; $k = 25$, 50%; $k = 30$, 49%; $k = 40$, 48%; $k = 50$, 43% (of 440 conditions). At $k = 20$ the per-network gap (A_{cond} motif $-$ matched) was $+0.026$ (median), positive in 7/10 networks; by motif type the per-condition win rate was line 138/263, saddle 32/76, point 34/101.

8.5 Held-out gains (prospective, seeds 0–2)

Estimating gains on one contiguous ring arc and predicting tangent directions at held-out angles gave $A_{\text{cond}} = 0.736$ vs. random 0.597 ($k = 20$); interleaved held-out $\approx$ in-sample (0.866 vs. 0.867); cross-task gains 0.812, cross-type gains 0.765.

9 Task accessibility of motif vs. matched non-motif modes (delay epoch; negative)

For each motif direction v and its eigenvalue-matched non-motif direction q (§8.4), evaluated at the fixed-point epoch, seven task-interface measures were compared (seeds 0–2, 128 conditions):

measure	motif	matched	motif > matched
input-drive projection $\lvert\cos(m, Bu)\rvert$	0.076	0.123	45/128
state-displacement projection	0.160	0.208	62/128
trajectory variance along mode	0.243	0.128	86/128
readout potency $\lVert R^T m\rVert^2$	0.001	0.001	53/128
persistence (local $\lvert\lambda\rvert$)	0.975	0.970	75/128
modal controllability $\lVert B^T w\rVert$	0.030	0.035	45/128
modal observability $\lVert C\, m\rVert$	0.101	0.121	55/128

$B = \alpha\,\mathrm{diag}(g)\,W_{\mathrm{in}}$ is the input Jacobian, C the response readout, R its orthonormal basis. Readout potency was ≈ 0.001 for both modes (chance ≈ 0.010).

10 Input-interpolation branch (re-found manifold; negative)

This is a negative result: the reduced quotient system does not reproduce the full network's re-found slow manifold as the input is interpolated. For a memory-ring task the input $u(\beta) = (1-\beta)u_{\mathrm{delay}} + \beta u_{\mathrm{response}}$ ramps the fixation channel from 1 to 0. At each β the slow set was re-found by relaxing a broad seed pool under the leaky dynamics; the full network and the reduced quotient system were compared. For `delaygo`, canonical network ($k = 20$): the full system had a one-dimensional slow set up to $\beta \approx 0.9$ while the reduced system had a single fixed point (effective dimension 0) throughout; marginal eigenvalue full 0.95–0.999, reduced 0.96–0.98; centroid cosine 0.74–0.98; slow-subspace principal angle 43°–63°; readout potency rose $0.000 \to 0.092$ (full) and $0.006 \to 0.026$ (reduced).

Sweep over k, task and seed. To show the failure is not specific to one setting the comparison was repeated over two tasks (`delaygo`, `delayanti`), two seeds (0 released, 3 corrected) and $k \in \{12, 20, 30\}$, averaging each metric over β.

seed	task	eff. dim (full)	eff. dim (reduced)	slow-plane angle	potency full / reduced
0	`delaygo`	0.73	0.00	52–58°	0.024 / 0.015
0	`delayanti`	1.20	0.00	50–53°	0.033 / 0.011
3	`delaygo`	1.64	0.00	73–78°	0.008 / 0.013
3	`delayanti`	2.32	0.00	77–79°	0.009 / 0.013

Values are averaged over β; the angle range spans $k \in \{12, 20, 30\}$. In all 12 cells the reduced effective dimension was 0 against a full effective dimension > 0.7. The reduced system's slow set collapsed (effective dimension near 0) while the full system's persisted, and the reduced slow plane stayed far from the full one, in every cell. The quotient captures the coarse existence and rough location of the memory manifold but not its dimensionality or fine geometry along the input interpolation, at any of the tested scales.

11 Rotation into the readout at the response epoch (Result 3, decisive control)

At the response epoch we compared the readout potency of the response-epoch ring plane (top-2 PCs of the response fixed-point cloud) with a matched non-ring plane (two eigenvectors of the response mean Jacobian with $\lvert\lambda\rvert$ nearest the ring modes', alignment < 0.40) and a random plane. Across the ten networks (109 delayed-response tasks), median readout potency: ring at the delay epoch

0.0004; ring at the response epoch 0.0302; matched non-ring 0.0015; random plane 0.0086. The ring plane exceeded the matched non-ring plane in 104/109 tasks (released seeds 33/33; retrained seeds 71/76) and exceeded the random plane in 100/109; per network the median ring potency exceeded the median matched potency in 10/10.

12 Concentration of motifs on quotient eigenmodes (Result 4)

This is a *within-network* statement: for each network separately, the collection of that network's task-epoch motifs is expressed in the coordinates of that network's own quotient, and we ask whether the motifs load a few of those modes rather than spreading over all k. No claim is made that a particular mode index recurs across networks — mode j of one network is not mode j of another — so all statistics are computed per network and only the resulting scalar is aggregated across seeds. The primary statistic is the fraction of aggregate motif energy on the top three modes; the secondary statistic is the participation ratio $\mathrm{PR} = (\sum_j E_j)^2 / \sum_j E_j^2$ (effective number of modes). Because the quotient is directed and non-normal its eigenvectors are not orthogonal, so the modes are orthonormalised (the lifted eigenvectors are Gram–Schmidt/QR ordered by $|\lambda|$) before energy is measured; §13 confirms that the low PR is not an artifact of this choice (a complex-aware orthonormal basis and the real Schur vectors give the same PR). The null is size-matched random partitions with their own quotient modes; significance is a two-sided Wilcoxon signed-rank test across the ten networks.

k	top-3 (conn)	top-3 (random)	conn > rand	Wilcoxon p	PR conn / rand
12	0.802	0.546	10/10	0.0020	2.72 / 7.03
20	0.641	0.430	10/10	0.0020	4.87 / 10.55
30	0.560	0.352	10/10	0.0020	6.59 / 14.70

The top-3 connectivity energy exceeded the random-quotient value in 10/10 networks at every k ($p = 0.0020$, the signed-rank floor for $n = 10$). Correlation of the top-3 connectivity energy with per-network task accuracy ($k = 20$, $n = 10$): Pearson $r = +0.47$ ($p = 0.17$), Spearman $\rho = +0.12$ ($p = 0.75$); the participation ratio versus accuracy: Pearson $r = -0.46$ ($p = 0.18$). Neither correlation is significant at $n = 10$.

12.1 W_{rec}-eigenbasis null (strong null; mixed)

Against the same W_{rec}-eigenbasis null used for containment (§7.1), the top-3 energy comparison is k-dependent rather than uniformly in favour of the connectivity quotient. Median top-3 energy on the connectivity quotient's top-3 eigenmodes versus the top-3 W_{rec} eigenvectors, over the ten networks:

k	top-3 (conn quotient)	top-3 (W_{rec}-eig)	Wilcoxon p
8	0.888	0.779	0.105
12	0.802	0.727	0.275
16	0.757	0.707	0.846
20	0.641	0.695	0.027
30	0.560	0.646	0.002
50	0.518	0.585	0.002

At small k ($k \leq 12$) the connectivity quotient concentrated motif energy on slightly fewer modes than the W_{rec} eigenbasis, but not significantly; at $k \geq 20$ the W_{rec} eigenbasis concentrated more

(significantly from $k = 30$). The concentration in §12 is thus specific relative to random quotients but is not sharper than the leading W_{rec} eigenvectors except at small k.

13 Mode-energy profiles by motif type

The concentration above pools all motif types. Here we ask, within each network, whether the three motif types occupy the *same* quotient modes or different ones, and we confirm the concentration is robust to the coordinate system. For each network we formed three aggregate mode-energy vectors — line attractors, point attractors, saddles — each an equal-weight mean of the per-motif energy profiles (every motif's profile normalised to sum one *before* pooling, so that the numerous line-attractor samples do not dominate; a per-task-family average gives the same conclusions). Energy is measured in an orthonormal basis of the quotient (below). All comparisons are within a network; only the resulting scalars are aggregated across the ten seeds.

Coordinate system (non-normality check). The quotient is directed and non-normal, so its eigenvectors are not orthogonal and raw-eigenvector energy/PR could be distorted. We therefore used two orthonormal bases whose leading columns span the dominant ($|\lambda|$-ordered) invariant subspace: (i) an eigen-ordered basis retaining both the real and imaginary parts of each complex conjugate pair (its full 2-D real invariant plane), Gram–Schmidt orthonormalised; and (ii) the real Schur vectors. The median per-motif participation ratio ($k = 20$) was 3.67 in the eigen-ordered orthonormal basis and 3.54 in the Schur basis, versus 3.33 for the earlier real-parts-only construction — i.e. the proper orthonormal bases give, if anything, a *higher* PR. The low participation ratio is therefore not an artifact of non-orthogonal eigenvectors. Within-type concentration held for all three types on their own network's modes (equal-weighted top-3 energy: saddles 0.72, lines 0.68, points 0.51; points least concentrated).

Do the types share modes? We compared the three type profiles pairwise by cosine and correlation, by top-3 mode-index overlap, and by the principal angles between the top-3 mode spans, each against size-matched random quotients (median over ten seeds; two-sided signed-rank of observed vs. null).

pair	cosine obs/null	p	top-3 overlap obs/null	p	angle° obs/null	p
line–point	0.767 / 0.795	0.77	0.67 / 0.37	0.037	30 / 57	0.037
line–saddle	0.813 / 0.730	1.00	0.50 / 0.35	0.049	45 / 59	0.049
point–saddle	0.792 / 0.756	0.92	0.67 / 0.36	0.020	30 / 58	0.020

The cosine (and correlation) of the full energy profiles did not differ from the random-quotient null for any pair ($p \geq 0.77$): all profiles place most of their energy on the single dominant mode, which inflates cosine similarity equally in the observed and null cases, so cosine is not discriminating here. The set- and subspace-based measures are: the three types shared more of their top-3 modes (0.50–0.67 vs. ≈ 0.36) and their top-3 spans were closer (30–45° vs. $\approx 58°$) than under random quotients, significantly for every pair ($p = 0.02$–0.05). Within a network the motif types thus occupy *overlapping* dominant mode-subspaces more than chance, rather than disjoint ones; the effect is visible in the top-mode overlap and principal angles but not in the profile cosine.

14 Cross-seed convergence of quotient modes

§12–13 are within-network. This section asks the across-network question directly: do independently trained seeds develop the *same* quotient modes? Modes cannot be matched by index or by lifted direction (different unit spaces, no correspondence), but the fifteen tasks are shared across seeds, so each mode is fingerprinted by a seed-invariant *task profile*: the 15-vector $T[j,r] = \sum_{m \in r} (p_m \cdot M_j)^2$ of how much each task's motifs load mode j (its eigenvalue $|\lambda|$ is also seed-invariant). For each seed we take the top-m modes by aggregate motif energy and L^1-normalise their task profiles. For every seed pair we Hungarian-match the two sets of modes by task-profile cosine and record the mean matched cosine; the null applies a random permutation of the 15-task axis to one seed's profiles (destroying task identity, preserving profile shape) and rematches.

At $k = 20$, top-6 modes: mean matched task-profile cosine 0.781 versus task-shuffled null 0.674, with observed $>$ null in 44/45 seed pairs (signed-rank $p < 10^{-4}$). The effect is robust across $k \in \{12, 20, 30\}$ and top-$m \in \{4, 6, 8\}$ (observed 0.75–0.80 vs. null 0.64–0.70; 43–45 of 45 pairs; $p < 10^{-4}$ throughout). The matched modes did *not* have similar eigenvalues (mean matched $|\lambda|$ gap ≈ 7.9): the convergence is in task function, not spectral magnitude. Consistently, the tasks dominating the top modes across all ten seeds were the delay / working-memory and context decision-making tasks (`contextdelaydm1` dominant for a top mode in 10 seeds, `delaydm1` in 9, `delaydm2` and `delayanti` in 7, `contextdelaydm2` in 6, `delaygo` and `dmsgo` in 5), and almost never the fast/reactive tasks (`fdgo`, `fdanti` ≤ 2). Independently trained networks therefore converge on a shared repertoire of quotient modes defined by which tasks recruit them — a memory/decision-dominated set — even though no individual mode index or eigenvalue corresponds across seeds. This is the earned cross-network version of the within-network concentration; it is a statement about task profiles, established by matching, not about mode ordering.

15 Eigenvalue (timescale) control

The motif directions are slow Jacobian modes ($|\lambda|$ near or above 1). A reviewer control: are containment, the quotient prediction A_{cond}, and the top-quotient-mode energy merely increasing functions of $|\lambda|$ (the implied timescale), so that "the motif lives in the connectivity subspace" would just restate "slow modes are captured"? For each condition we took *all* eigenvectors of the mean Jacobian $\bar{J}$ (not only the motif direction) and recorded, per eigenvector, its containment $\|P_H v\|$, A_{cond} (against the same $r = 3$ predicted-quotient basis), and its top-3 quotient-mode energy, together with $|\lambda|$ and the timescale $\tau = -1/\log|\lambda|$. Pooling all 94,237 eigenvectors over the ten networks ($k = 20$):

	vs $\lvert\lambda\rvert$: r (R^2)	vs $\log\tau$: r	shape across $\lvert\lambda\rvert$
containment	-0.05 (0.002)	-0.00	flat 0.27 in bulk, rises to 0.68 for $\lvert\lambda\rvert > 1.05$
A_{cond}	$+0.35$ (0.12)	$+0.28$	$0.35 \to 0.84 \to 0.94$ across the slow tail
top-3 energy	$+0.07$ (0.004)	$+0.01$	flat 0.07 in bulk, rises to 0.73 for $\lvert\lambda\rvert > 1.05$

Globally none of the three is a strong linear function of $|\lambda|$ ($R^2 \leq 0.12$), because slow modes ($|\lambda| \gtrsim 0.95$) are only $\approx 3\%$ of eigenvectors; but within that slow tail all three rise steeply. The fair test is therefore a matched-$|\lambda|$ comparison: for each motif eigenvector, the quantity minus its mean over non-motif eigenvectors of the *same* $|\lambda|$ (window ± 0.02, per network), aggregated across seeds:

quantity	motif − matched non-motif	seeds > 0	signed-rank p
containment	+0.022	6/10	0.43
A_{cond}	+0.085	9/10	0.004
top-3 mode energy	+0.199	10/10	0.002

The results dissociate. *Containment is a timescale phenomenon*: at matched $|\lambda|$ the motif directions are contained no more than other Jacobian eigenvectors (+0.02, $p = 0.43$) — slow/near-marginal modes generally lie in the connectivity subspace, and the motifs inherit this by being slow. (A naive residual against the near-flat global $|\lambda|$–containment fit gives +0.39, 10/10, $p = 0.002$, but that overstates the effect precisely because the global fit ignores the slow-tail rise; the matched control is the correct test and it is null.) By contrast the quotient prediction and the concentration are *not* explained by timescale: at matched $|\lambda|$ the motif directions are still better predicted by the reduced quotient (A_{cond} +0.085, $p = 0.004$) and far more concentrated on the top quotient modes (+0.199, $p = 0.002$) than non-motif modes of the same $|\lambda|$. Thus §7 should be read as largely a statement about slow modes, whereas §8 and §12 survive the timescale control.

16 Activation robustness: released tanh networks

The main analyses use `softplus` networks. To test activation dependence we ran the same pipeline on the three released `tanh` diagonal-init 256-unit networks (`LeakyRNN_tanh_diag_256_all_0..2`): active units by the same $> 10^{-3}$ variance criterion (all 256 units are active — `tanh` does not sparsify as `softplus` does), the signed-WL partition of the transpose-variant W_{rec} at the operating scale ($k = 51$), fixed-point finding across the task epochs, and the mode concentration. We compare to the three *released* `softplus` seeds (0–2) so the sample size ($n = 3$) and test are matched.

Lumpability (Result 1): holds. Eigenvector-leakage z of W_{rec} onto the group-constant subspace vs. size-matched random partitions was strongly negative in all three `tanh` networks: transpose-orientation leakage $z = -17.7, -17.1, -19.8$ (invariance $z = -27, -23, -28$; original-orientation leakage $z = -8.0, -6.3, -8.4$). These are in the same range as the `softplus` networks (§3). The signed connectivity partition is lumpable under `tanh` as well.

Motif inventory differs. The `tanh` fixed-point inventory is *point-attractor dominated* (18/11/6, 18/5/0, 22/5/2 point/line/saddle per seed; 58 points, 21 lines, 8 saddles over the three nets), unlike the line-attractor–dominated `softplus` inventory (§6).

Concentration (Result 4): does not hold. Motif energy did *not* concentrate on the connectivity quotient's modes beyond a random quotient in the `tanh` networks, in contrast to `softplus` on the matched three released seeds:

activation	k	PR conn / rand (z)	top-3 conn / rand	median per-net perm. p
`softplus` (0–2)	12	2.85 / 6.90 (−2.8)	0.785 / 0.551	0.003
	20	3.67 / 10.64 (−3.2)	0.797 / 0.426	0.003
	30	5.04 / 15.36 (−3.6)	0.670 / 0.337	0.003
`tanh` (0–2)	12	5.88 / 6.29 (−0.3)	0.587 / 0.586	0.37
	20	10.36 / 10.55 (+0.7)	0.437 / 0.428	0.71
	30	19.23 / 14.22 (+1.4)	0.264 / 0.350	0.92

At matched n and with the identical test, the `softplus` motifs concentrated far below their own random-quotient null (per-network $z \approx -3$; top-3 energy ≈ 0.8 vs. 0.4), while the `tanh` motifs did not (per-network $z \approx 0$; top-3 energy at the random level). By type at $k = 20$ the only hint of `tanh` concentration was in the few saddles ($z \approx -1.1$, $n = 6$); the dominant point attractors did not concentrate ($z \approx +0.6$). Thus the lumpability of the connectivity partition is activation-general, but the localisation of fixed-point motifs on the quotient's dominant modes is specific to the (line-attractor–dominated) `softplus` networks and does not transfer to the (point-attractor–dominated) `tanh` networks. The three tanh networks are a small sample and this is reported as a robustness check, not a seed-level significance test.

17 Cross-task participation reuse and its control (negative)

For each motif a participation vector over the groups (summed squared loading) was formed; the reuse statistic R is the mean same-computation correlation minus the mean different-computation correlation. Against a size-matched random partition (each minus its own baseline), across the ten networks:

k	R (conn)	R (random)	R (activity)	conn > rand	p
8	0.127	0.195	0.191	0/10	0.002
12	0.123	0.193	0.180	2/10	0.010
20	0.147	0.197	0.180	3/10	0.193
30	0.158	0.194	0.195	1/10	0.020
50	0.194	0.197	0.201	6/10	0.695

The connectivity participation reuse did not exceed the size-matched random partition at any k (median $R_{\text{conn}} < R_{\text{random}}$ throughout; connectivity above random in 0–6 of 10 networks depending on k, never with $p < 0.05$ in its favour).

17.1 Relation to the concentration result (§12)

Because connectivity motif energy concentrates on few quotient modes (§12), one might expect the reuse deficit here to be a mechanical consequence: concentrated participation vectors are supported on fewer groups, which could compress the variance of between-motif correlations and lower R. We tested this directly at $k = 20$ using the reuse statistic above (cross-task pairs, group-projected profiles). The connectivity participation vectors were indeed more concentrated than random — median effective group support 4.9–7.7 versus 6.5–10.5, lower in 10/10 networks — but the reuse deficit was null-sized ($\overline{R_{\text{conn}} - R_{\text{random}}} = -0.013$) and did *not* track concentration across seeds (Spearman $\rho = -0.03$, $p = 0.93$ between the per-seed support gap and reuse gap). The two observations are therefore reported as separate findings: concentration is a robust positive against random quotients (§12); the reuse deficit is a small, essentially null-sized negative (this section) that concentration does not explain.

18 Trajectory decoding from the quotient coordinates

Does the k-dimensional connectivity quotient carry the computation, or does group-averaging discard the task-relevant state? For each network we ran the fifteen tasks (noise off) and, at every retained epoch, took the instantaneous hidden state at four sampled times. We decoded two trajectory variables — task identity (15-way) and computational phase (fixation / stimulus / delay / response,

pooled across tasks) — with balanced multinomial logistic regression (5-fold CV) from four matched feature sets: the full active state (≈ 210 dims, the ceiling), the WL role-group means (k dims), a size-matched random-partition's group means (k), and a random Gaussian projection to k dims. Mean CV accuracy over the ten networks ($k = 20$):

variable	full	WL quotient	random partition	random projection	chance	WL vs rand-proj
task identity	1.000	0.727	0.750	0.733	0.223	5/10, $p = 0.62$
computational phase	0.997	0.889	0.812	0.791	0.311	10/10, $p = 0.002$

The two variables dissociate. Computational phase was decoded from the k WL quotient coordinates better than from either k-dimensional random baseline (10/10 networks, signed-rank $p = 0.002$ against both), i.e. the coarse connectivity coordinates preferentially retain the dynamical stage of the trajectory. Task identity was not: the WL quotient decoded it no better than a random projection of the same dimension (5/10, $p = 0.62$). This is consistent with the quotient capturing the slow dynamical state (which carries the phase) rather than the input-driven task label (which the full state decodes at ceiling but the slow subspace does not privilege).

19 Causal block-model resample (negative)

The results above are correlational: they measure how far the quotient coordinates track the trained dynamics. This is a causal test of whether preserving the quotient Q_{struct} is sufficient for behaviour. We resampled the trained recurrent matrix under two surrogates sharing the same global weight multiset but differing in whether the block structure is kept, injected each back into the network with all other weights fixed and no retraining, and measured mean performance on the fifteen tasks:

- *Q-preserving*: permute residuals within each (g_i, g_j) role block. This preserves every block mean, hence $Q_{\text{struct}} = D^{-1}H^T W_a H$ exactly, while destroying within-block wiring.
- *partition-agnostic*: permute all active–active entries globally. Same weight multiset, but block means (and Q_{struct}) destroyed.

Means over the ten 256-unit networks ($k = 20$, five surrogate draws each); performance is the mean fraction correct over the fifteen tasks, the spectral error is the mean $|\Delta\lambda|$ of the top-ten $|\lambda|$ of the active recurrent matrix relative to the trained one.

	trained	Q-preserving	partition-agnostic
task performance	0.944	0.044	0.042
top-10 $\lvert\lambda\rvert$ error	—	6.16	9.28

Both surrogates collapsed performance from the trained ceiling (0.944) to near the fixation floor (≈ 0.04), and the Q-preserving surrogate was *not* better than the partition-agnostic one (Q-preserving above partition-agnostic in 6/10 networks, signed-rank $p = 0.62$). Preserving the coarse 20-group quotient (a few hundred block means out of $\sim 4 \times 10^4$ weights) is thus not causally sufficient for the computation: the fine within-block wiring carries the behaviour. However, the Q-preserving surrogate did preserve the trained matrix's leading eigenvalues markedly better than the partition-agnostic shuffle (spectral error 6.16 vs. 9.28; Q-preserving lower in 10/10 networks, $p = 0.002$). The two readouts dissociate: preserving Q_{struct} retains the slow eigenspectrum — the structure the correlational results (§3–18) concern — but not enough of the fine wiring to retain behaviour. This bounds the quotient claim: the connectivity coordinates are a real property of the slow dynamical subspace, not a sufficient causal substitute for the trained weights.

20 Statistical conventions

Cross-network claims treated the seed as the unit of replication; effects were tested across seeds with a two-sided Wilcoxon signed-rank test ($p = 0.002$ is the floor for $n = 10$). Within-network nulls preserved group sizes and permuted unit labels (or, for similarity analyses, unit identities); each surrogate or size-matched partition was scored against its own recomputed null. Standardised scores are relative to the null mean and standard deviation, oriented positive-for-more-structure. Permutation nulls used 1000 draws (a first pass at 200 gave the same signs and comparable magnitudes).

Multiple comparisons. The report contains one confirmatory family and a set of control/exploratory tests. The confirmatory family is the four backbone results, each as a single across-seed signed-rank test at the operating scale: (R1) lumpability leakage, (R2) quotient prediction > random, (R3) rotation ring > matched, (R4) top-3 concentration > random. Four tests: at Bonferroni $\alpha = 0.05/4 = 0.0125$ all four remain significant (R1, R2, R4 at the $p = 0.002$ floor; R3 by the 104/109 binomial). The k-sweep entries for a given result are the same hypothesis re-tested at neighbouring scales, not independent members of the family, and are reported as a robustness curve rather than counted separately. The negative controls (eigenvalue-matched, task-accessibility, cross-task reuse), the strong-null comparisons (W_{rec}-eigenbasis, input-weight span), the surrogate mechanism, and the accuracy correlations are diagnostic and are reported at face value without family correction; none is used to support a positive claim.

21 Data and code provenance

Analyses were run under the `flexsym` package. The result tables and JSON summaries are:

- `lumpability_ksweep.csv`, `lumpability.csv` (§3);
- `training_control_all_sizes.csv` (§3.2);
- `quotient_prediction.json`, `quotient_controls.json` (§8);
- `quotient_eigmatched_control.json` (§8.4);
- `subspace_role_vs_activity.json` (§7);
- `rotation_readout_control.csv` (§11);
- `quotient_mode_concentration.json` (§12);
- `task_accessibility.csv` (§9);
- `reuse_random_partition_control.json` (§17);
- `motif_taxonomy_all_nets.csv` (§6).

The uncorrected first-batch models and superseded outputs are documented in `legacy_uncorrected_seeds_report`.

A Per-seed tables

All tables are the ten 256-unit networks (seeds 0–2 released, 3–9 retrained on the corrected objective) unless noted. "score" denotes a signed standardised deviation from a size-matched null (positive-for-more-structure).

A.1 Lumpability, per seed (leakage and invariance scores at $k = 8, 20, 50$; §3)

		leak score (W^T)			invariance score		
seed	model	k=8	k=20	k=50	k=8	k=20	k=50
0	released	+32.1	+27.4	+16.6	−2.7	+1.8	+17.6
1	released	+51.2	+27.3	+19.2	−12.1	−1.6	+11.1
2	released	+28.8	+27.3	+15.8	+1.1	+3.8	+21.4
3	retrained	+14.0	+17.9	+17.9	+1.1	+5.0	+19.8
4	retrained	+25.1	+28.9	+19.8	+1.1	+9.4	+17.8
5	retrained	+27.5	+28.1	+20.9	+1.1	+9.6	+15.3
6	retrained	+37.7	+29.9	+16.8	−0.4	+7.8	+18.8
7	retrained	+28.0	+31.2	+19.5	+0.9	+10.4	+15.8
8	retrained	+21.0	+23.0	+20.3	+1.1	+8.1	+21.9
9	retrained	+20.3	+30.5	+20.8	+1.2	+6.2	+16.5

A.2 Degree/training surrogate control, per network (§3.2)

256-unit seeds 0–9 (corrected); 128/512/1024-unit seeds 0–2 (released).

		trained		row shuffle		full shuffle		Gaussian	
size	seed	eq	R^2	eq	R^2	eq	R^2	eq	R^2
128	0	+3.1	0.753	+12.1	0.778	+0.7	0.385	+3.3	0.338
128	1	+5.5	0.816	+11.5	0.799	+2.8	0.428	+1.7	0.384
128	2	+4.4	0.825	+12.6	0.786	+3.1	0.426	+3.7	0.278
256	0	+4.2	0.831	+14.8	0.827	+1.5	0.455	+2.4	0.333
256	1	+5.7	0.890	+10.5	0.577	+2.6	0.355	+3.5	0.245
256	2	+7.5	0.834	+16.0	0.753	+0.5	0.411	+2.1	0.279
256	3	+8.7	0.727	+13.7	0.681	+4.5	0.433	+2.8	0.260
256	4	+7.2	0.847	+8.1	0.677	+3.1	0.352	+2.5	0.301
256	5	+9.0	0.673	+15.1	0.643	+5.5	0.476	+1.8	0.254
256	6	+7.4	0.756	+10.3	0.679	+4.3	0.368	+2.4	0.227
256	7	+7.0	0.750	+11.1	0.722	+4.4	0.434	+1.6	0.279
256	8	+6.9	0.746	+10.3	0.725	+2.9	0.406	−0.5	0.305
256	9	+10.0	0.787	+12.7	0.649	+2.1	0.459	+2.6	0.268
512	0	+6.0	0.839	+17.3	0.748	+1.9	0.288	+4.3	0.274
512	1	+1.7	0.844	+18.0	0.777	+2.4	0.336	+2.8	0.294
512	2	+5.3	0.857	+19.5	0.818	+3.1	0.398	+1.7	0.296
1024	0	−4.2	0.835	+17.0	0.769	+1.8	0.348	+1.9	0.218
1024	1	−6.2	0.851	+18.8	0.778	+1.2	0.336	+3.6	0.219
1024	2	+0.6	0.878	+20.3	0.843	+3.0	0.398	+2.0	0.275

A.3 Quotient prediction, per seed ($k = 20$, $r = 3$; §8)

seed	model	A_{cond}	rand	activity	z	$\lvert\Delta\lambda\rvert$ conn	rand
0	released	0.889	0.642	0.703	+4.02	0.047	0.193
1	released	0.896	0.627	0.626	+4.19	0.031	0.195
2	released	0.846	0.637	0.646	+3.65	0.043	0.180
3	retrained	0.807	0.612	0.603	+3.10	0.056	0.187
4	retrained	0.821	0.584	0.658	+3.81	0.038	0.188
5	retrained	0.872	0.603	0.625	+4.27	0.046	0.193
6	retrained	0.774	0.614	0.547	+2.41	0.042	0.189
7	retrained	0.791	0.589	0.621	+3.25	0.048	0.198
8	retrained	0.811	0.610	0.584	+3.03	0.046	0.188
9	retrained	0.758	0.613	0.685	+2.26	0.076	0.202

A.4 Motif-direction containment, per seed ($k = 20$; §7)

seed	model	#motifs	Δ_{role}	Δ_{activity}	z (role−act)
0	released	47	0.329	−0.000	+8.84
1	released	33	0.322	0.041	+8.38
2	released	48	0.285	0.067	+6.33
3	retrained	44	0.210	0.022	+5.09
4	retrained	43	0.243	−0.004	+7.30
5	retrained	42	0.205	−0.007	+5.42
6	retrained	48	0.335	0.016	+7.31
7	retrained	43	0.211	0.047	+5.18
8	retrained	45	0.280	−0.010	+8.45
9	retrained	47	0.279	0.007	+7.57

A.5 Rotation into the readout, per seed (§11)

Median readout potency over each network's delayed-response tasks.

seed	model	tasks	ring (delay)	ring (resp.)	matched	random
0	released	11	0.0000	0.1014	0.0058	0.0098
1	released	11	0.0001	0.1007	0.0018	0.0081
2	released	11	0.0000	0.0631	0.0041	0.0096
3	retrained	11	0.0009	0.0288	0.0006	0.0079
4	retrained	10	0.0007	0.0196	0.0020	0.0086
5	retrained	11	0.0003	0.0293	0.0004	0.0088
6	retrained	11	0.0002	0.0291	0.0045	0.0080
7	retrained	11	0.0127	0.0347	0.0004	0.0079
8	retrained	11	0.0006	0.0427	0.0009	0.0082
9	retrained	11	0.0008	0.0197	0.0015	0.0081

Ring-response > matched (per task): all tasks for seeds $0, 1, 2, 3, 5, 6, 8, 9$; 9/10 (seed 4); 7/11 (seed 7).

A.6 Mode concentration, per seed (top-3 energy and PR; §12)

		PR k=12		PR k=20		PR k=30		top-3 (k=20)	
seed	model	conn	rand	conn	rand	conn	rand	conn	rand
0	released	4.04	6.60	4.13	10.27	5.37	13.88	0.80	0.44
1	released	2.85	7.83	3.37	11.45	4.45	15.89	0.80	0.41
2	released	2.52	6.90	3.67	10.64	5.04	15.36	0.67	0.43
3	retrained	2.17	7.27	4.60	10.81	7.81	15.26	0.60	0.42
4	retrained	4.23	6.45	6.59	10.46	9.26	13.77	0.60	0.43
5	retrained	2.40	6.79	5.15	9.22	7.01	13.16	0.63	0.47
6	retrained	2.18	5.40	6.30	8.66	6.42	13.01	0.54	0.49
7	retrained	4.03	7.60	6.13	9.35	8.02	14.14	0.64	0.46
8	retrained	2.77	7.16	4.40	12.08	6.65	16.88	0.70	0.39
9	retrained	2.66	7.59	5.45	11.53	6.54	16.18	0.64	0.40

Per-seed task accuracy (for the accuracy correlation, §12): seeds 0–9 = 0.991, 0.974, 0.987, 0.975, 0.976, 0.974, 0.974, 0.9

A.7 Eigenvalue-matched non-motif control, per seed ($k = 20$; §8.4)

seed	model	#cond	motif	matched	win
0	released	47	0.889	0.914	15/47
1	released	33	0.896	0.744	18/33
2	released	48	0.846	0.806	24/48
3	retrained	44	0.807	0.709	26/44
4	retrained	43	0.821	0.765	23/43
5	retrained	42	0.872	0.861	13/42
6	retrained	48	0.774	0.792	23/48
7	retrained	43	0.791	0.820	17/43
8	retrained	45	0.811	0.746	25/45
9	retrained	47	0.758	0.752	20/47

A.8 Cross-task participation reuse, per seed ($k = 20$; §17)

seed	model	R conn	R random	R activity	Δ
0	released	0.305	0.256	0.272	+0.048
1	released	0.136	0.141	0.155	−0.005
2	released	0.129	0.176	0.193	−0.046
3	retrained	0.128	0.185	0.115	−0.058
4	retrained	0.141	0.221	0.148	−0.080
5	retrained	0.140	0.205	0.167	−0.065
6	retrained	0.202	0.191	0.167	+0.011
7	retrained	0.238	0.202	0.193	+0.036
8	retrained	0.153	0.189	0.217	−0.036
9	retrained	0.200	0.206	0.225	−0.006

$\Delta > 0$ in 2/10 networks (seeds 6, 7; and seed 0).

A.9 Task accessibility, per seed (seeds 0–2; §9)

Number of conditions with motif mode > matched mode, per measure.

seed	#cond	input	displ.	traj. var	readout	persist.	controll.	observ.
0	47	13/47	19/47	33/47	22/47	32/47	19/47	23/47
1	33	9/33	13/33	19/33	18/33	12/33	5/33	18/33
2	48	23/48	30/48	34/48	13/48	31/48	21/48	14/48

A.10 Fixed-point taxonomy, per seed (corrected models; §6)

	gate (matched / total)			inventory (#epochs)				
seed	mem→line	dec→saddle	sens→point	line	point	saddle	mixed	rot.
0	16/16	14/14	4/4	26	12	9	2	0
1	8/16	3/14	1/4	16	14	3	16	0
2	16/16	13/14	4/4	25	15	8	1	0
3	10/16	12/14	1/4	27	10	7	5	0
4	13/16	13/14	1/4	27	6	10	6	0
5	13/16	10/14	0/4	26	8	8	7	0
6	16/16	7/14	1/4	32	11	5	1	0
7	12/16	12/14	1/4	27	9	7	6	0
8	12/16	13/14	0/4	27	9	9	4	0
9	14/16	12/14	0/4	30	7	10	2	0